

Microeconomic Analysis: Problem Set VI

Due to week 7: submit problems 1, 2 and 6

1 EU or not?

Determine whether the following preferences (decision criteria) are consistent with the EU hypothesis (Ariel Rubinstein's examples). Assume $L = [p_1, \dots, p_N; x_1, \dots, x_N]$ with $x_1 < \dots < x_N$

- (a) Preference for "less dispersion" $U(L) = -\sum (p_i - \frac{1}{N})^2$.
- (b) Preference for "more certainty" $U(L) = \max\{p_1, \dots, p_N\}$.
- (c) More likely "good outcomes" $U(L) = \sum_{i \geq G} p_i$ for a given G
- (d) Better "worst-case" $U(L) = \min\{u(x_i) \text{ such that } p_i > 0\}$
- (e) Better "most-likely prize" $U(L) = \{\text{highest } k \text{ with } p_k \geq p_i, \forall i\}$

2 Risk Sharing

Lottery L pays $x_k \in \mathbb{R}$ with probability p_k , where $k \in \{1, \dots, K\}$. There are N agents with identical, strictly risk averse EU preferences on final wealth. Every agent has the same initial wealth, w .

Consider two possible arrangements. In the first arrangement, an agent is picked at random (each with probability $1/N$); then the outcome of the lottery is given to the selected agent while all other agents receive nothing. In the second arrangement agents share the outcome of the lottery equally. Which arrangement do the agents prefer?

Provide a mathematical proof that relies on the strict concavity of the agents' Bernoulli utility function.

3 PhD or not?

Ann has wealth worth £1 million (£1m, mainly her human capital). If she pursues a PhD then, she believes, she will end up with £4m with 10% chance (a star in academia) or with £0.8m with 90% chance (five wasted years). She has expected-utility preferences over her wealth, and she is indifferent between doing a PhD and not.

- (a) Set her utility of £0.8m to 0 and that of £4m to 1; what is her utility of £1m?
- (b) Is Ann risk neutral, risk averse, risk loving, or cannot tell?
- (c) A career advisor tells Ann that even without a PhD she has a 6% chance of ending up with £4m, a 30% chance of falling to £0.8m, and a 64% chance of remaining at £1m. Does this first- or second-order stochastically dominate doing a PhD, or vice versa?
- (d) Assume the career advisor is right. Furthermore, assume that earning a PhD eliminates the possibility that Ann ends up with £0.8m wealth (only £4m or £1m are possible). How likely should be the best outcome to convince Ann that a PhD is worth doing?

4 Prize Choice (due to Osborne and Rubinstein)

An individual has expected utility preferences. The individual has wealth w and has to choose an amount $x \leq w$, after which a lottery is conducted in which with probability α she gains $2x$ and with probability $(1 - \alpha)$ she loses x .

- (a) Show that, if the individual is risk averse, the higher the value of α , the higher the amount that she chooses.
- (b) Suppose the individual has CRRA(r) preferences with $r \in (0, 1)$. Compute the amount chosen and show that it decreases with the level of risk aversion of the individual. Check her optimal choice with CRRA(1), i.e., logarithmic preferences.

5 EV and CE

Consider lotteries, all of them with payoffs in $[0, 1]$, that have density functions of the form $f(z) = (\beta + 1)z^\beta$, with $\beta \in [0, 1]$ (notice that $\beta = 0$ corresponds to the uniform distribution).

- (a) Show that the expected value of the lotteries is increasing in β . Can you easily conclude from there that the best lottery is that with $\beta = 1$? Can you maybe use an argument based on first order stochastic dominance?
- (b) An individual has risk averse CRRA preferences z^{1-r} , with a risk aversion parameter $r \in [0, 1]$. Compute the Expected Utility and the Certainty Equivalent of each lottery.

6 Investment

A risk-averse investor with a strictly increasing Bernoulli utility function u has £1 initial wealth that she has to invest. Two risky assets are available to her; their gross returns, per £invested, are X and Y . If she invests α proportion of her wealth in X and $(1 - \alpha)$ proportion of it in Y then her final wealth is $\alpha X + (1 - \alpha)Y$. The random variables X and Y are independent and have the same mean, though their distributions (cdfs) may be different.

- (a) Denote the amount invested in the two assets by α and $(1 - \alpha)$, respectively. Write down the agent's expected utility, $V(\alpha)$. Show that the optimal value of α satisfies $V'(\alpha) = \mathbb{E}[(X - Y)u'(\alpha X + (1 - \alpha)Y)] = 0$.
- (b) Show that the investor invest a positive amount in each asset: $\alpha \in (0, 1)$. You may use the fact that for any non-degenerate real-valued random variable Z and any strictly decreasing function g , we have $\mathbb{E}[Zg(Z)] < \mathbb{E}[Z]\mathbb{E}[g(Z)]$. (This inequality holds because the covariance of Z and $g(Z)$ is negative.)